

Harish Chinnakadiri
AI/ML Engineer | Gen AI Engineer
+12549105395 | chinnakadiriharish@gmail.com |

SUMMARY

Results-driven AI/ML & Generative AI Engineer with 4+ years of experience developing machine learning, NLP, and LLM-powered solutions across healthcare and banking/financial services domains. Skilled in Python, TensorFlow, XGBoost, NLP/LLMs, RAG, LangChain, LangGraph, Amazon Bedrock, SageMaker, and MLOps, with hands-on experience building predictive models, intelligent retrieval systems, document-processing workflows, and scalable AI services.

SKILLS

Programming & Frameworks: Python, SQL, PySpark, TensorFlow, Keras, scikit-learn, Pandas, NumPy

Machine Learning & AI: Supervised Learning, Logistic Regression, Random Forest, XGBoost, Neural Networks, Predictive Modeling, Feature Engineering, Hyperparameter Tuning, MLflow

NLP & Generative AI: Generative AI, LLMs, RAG, Agentic AI, BERT, spaCy, LangChain, LangGraph, Hugging Face, Prompt Engineering, Embeddings, Semantic Search, Guardrails

LLM Platforms: Meta Llama, Amazon Bedrock

Cloud & Containers: AWS SageMaker, Amazon S3, Amazon EKS, AWS Lambda, Docker

Big Data & Databases: PySpark, PostgreSQL, pgvector

MLOps & Monitoring: GitHub Actions, CI/CD, Terraform, Amazon CloudWatch, Drift Detection

APIs & Tools: FastAPI, REST APIs, Git, Postman, Swagger/OpenAPI, SHAP, LIME

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Humana

May 2025 – Present | USA

- Developed **Generative AI and RAG workflows** using **Python, LangChain, Amazon Bedrock, and PostgreSQL/pgvector** for healthcare document search and summarization, improving retrieval relevance by approximately **22%** through chunking and metadata-filtering optimization.
- Integrated **Meta Llama** models through **Amazon Bedrock** to generate context-grounded responses from retrieved healthcare information, applying **Prompt Engineering** and response-validation techniques to improve answer consistency.
- Implemented **semantic search and embedding pipelines** for healthcare documents, experimenting with chunk sizes, similarity thresholds, and top-k retrieval strategies to improve the quality of retrieved context.
- Developed workflow orchestration using **LangGraph** for multi-step retrieval and document-processing tasks, contributing to controlled tool integrations with approved internal services.
- Built **REST API endpoints using FastAPI** to expose document-search and summarization capabilities to downstream applications, reducing average API response latency from approximately **300ms to 220ms** through request and retrieval optimization.
- Supported containerization, deployment, and monitoring** using **Docker, Amazon EKS, SageMaker, and CloudWatch**, collaborating with platform engineers to troubleshoot application issues and maintain reliable AI services.

Data Scientist, Wells Fargo

May 2020 – Jul 2023 | USA

- Performed **exploratory data analysis and preprocessing** using **Python, SQL, Pandas, and NumPy** across customer and transaction datasets, identifying missing values, outliers, class imbalance, and data-quality issues.
- Built **machine learning models** using **Logistic Regression, Random Forest, and XGBoost** for customer and transaction risk analysis, improving baseline predictive performance by approximately **15%** through feature engineering and tuning.
- Evaluated **predictive models** using **precision, recall, F1-score, ROC-AUC, and cross-validation**, applying hyperparameter optimization to improve model generalization across customer segments.
- Developed **NLP workflows** using **spaCy** and later **BERT-based representations** for financial-document classification and downstream analytical workflows.
- Assisted with model training and experimentation using **Amazon SageMaker, S3, and MLflow**, maintaining model parameters, evaluation metrics, and artifacts across iterative development cycles.
- Implemented model explainability using **SHAP** and supported **Docker-based** inference deployment and **CloudWatch** monitoring while collaborating with senior team members on model validation and production troubleshooting.

EDUCATION

Master of Science in Data Science, University of North Texas,
Denton, Tx

Aug 2023 – May 2025

Bachelor's Degree, Vellore Institute of Technology (VIT),
India

Jul 2017 – May 2021

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty
- Machine Learning Specialization – Coursera
- Oracle AI Vector Search Professional
- AWS Certified Data Analytics – Specialty